

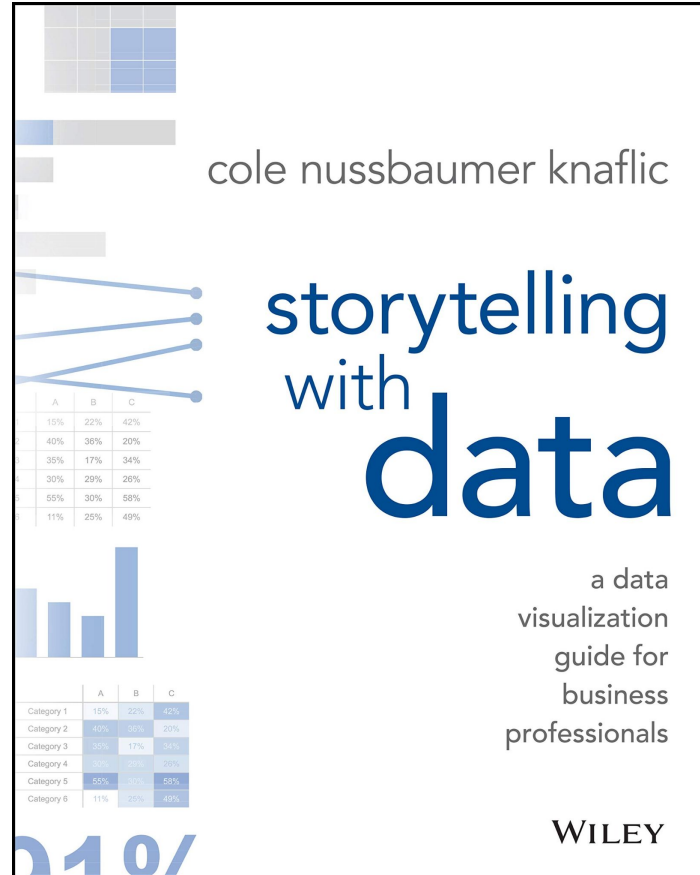
---

# Data Visualization

Rennan Marujo



# Core book





---

# The importance of context

# Context

What background information is relevant or essential?

- Who is the audience or decision maker? What do we know about them?
- What biases does our audience have that might make them supportive of or resistant to our message?
- What data is available that would strengthen our case? Is our audience familiar with this data, or is it new?
- Where are the risks: what factors could weaken our case and do we need to proactively address them?
  - What would a successful outcome look like?
- If you only had a limited amount of time or a single sentence to tell your audience what they need to know, what would you say?



—  
**choosing an  
effective  
visual**

# Choosing an effective visual

91%

Simple text

|            | A   | B   | C   |
|------------|-----|-----|-----|
| Category 1 | 15% | 22% | 42% |
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| Category 4 | 30% | 29% | 26% |
| Category 5 | 55% | 30% | 58% |
| Category 6 | 11% | 25% | 49% |

Table

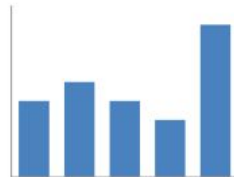
|            | A   | B   | C   |
|------------|-----|-----|-----|
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Heatmap

FIGURE 2.1 The visuals I use most



Scatterplot



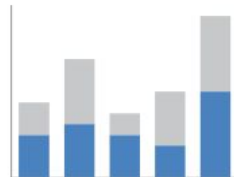
Vertical bar



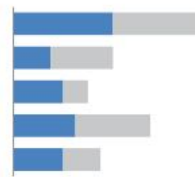
Horizontal bar



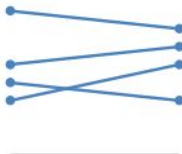
Line



Stacked vertical bar



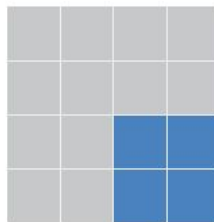
Stacked horizontal bar



Slopegraph



Waterfall



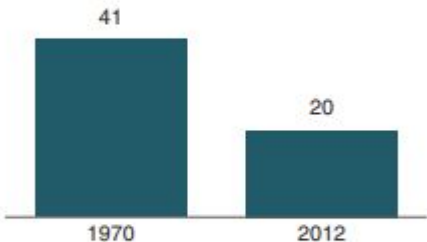
Square area

# Choosing an effective visual

## Simple text

### Children with a "Traditional" Stay-at-Home Mother

*% of children with a married  
stay-at-home mother with a  
working husband*



Note: Based on children younger than 18. Their mothers are categorized based on employment status in 1970 and 2012.

Source: Pew Research Center analysis of March Current Population Surveys Integrated Public Use Microdata Series (IPUMS-CPS), 1971 and 2013

Adapted from **PEW RESEARCH CENTER**

# 20%

of children had a  
**traditional stay-at-home mom**  
in 2012, compared to 41% in 1970

# Choosing an effective visual

## Simple text



Vertical bar



# Choosing an effective visual

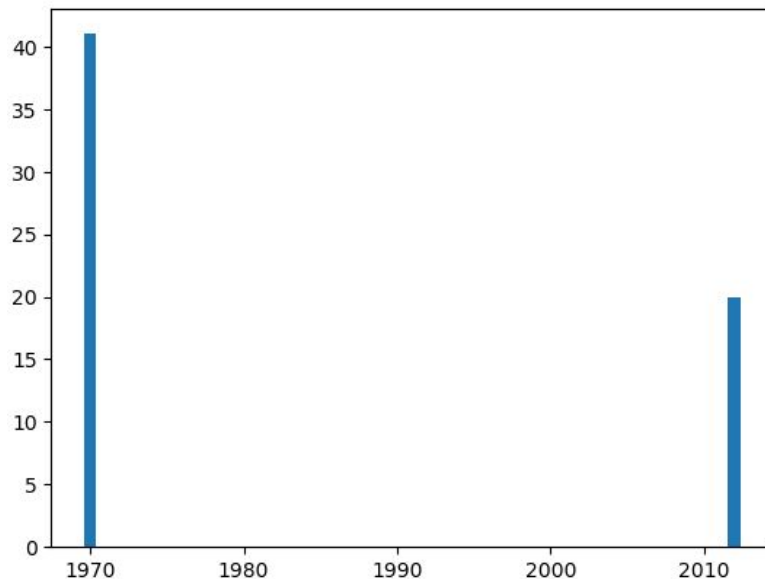
## Simple text

```
import matplotlib.pyplot as plt

years = [1970, 2012]
percentages = [41, 20]

plt.bar(years, percentages)

plt.show()
```



# Choosing an effective visual

## Simple text

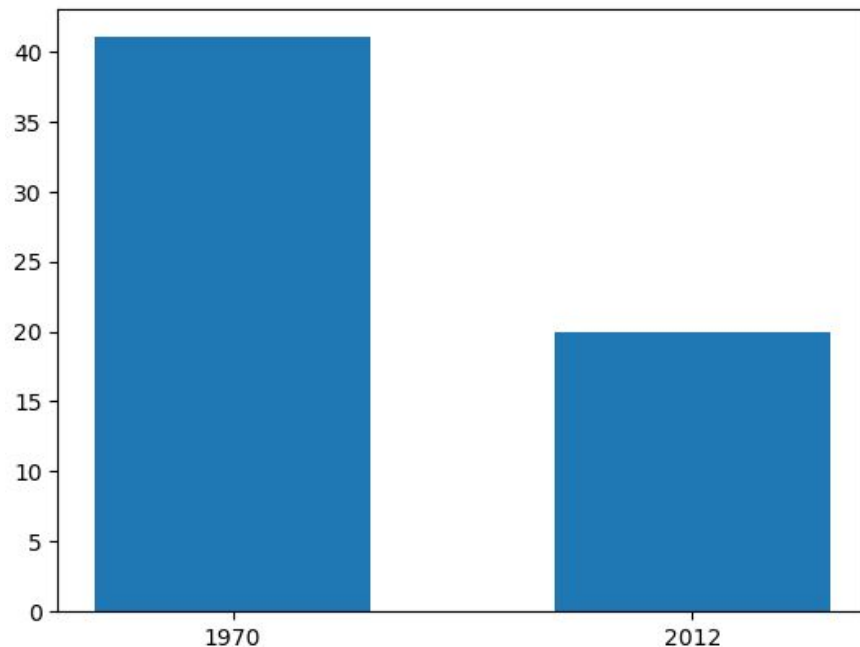
```
import matplotlib.pyplot as plt

years = [1970, 2012]
percentages = [41, 20]

x = [0, 1]

plt.bar(x, percentages, width=0.6)
plt.xticks(x, years)

plt.show()
```



# Choosing an effective visual Tables

Heavy borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

Light borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

Minimal borders

| Group   | Metric A | Metric B | Metric C |
|---------|----------|----------|----------|
| Group 1 | \$X.X    | Y%       | Z,ZZZ    |
| Group 2 | \$X.X    | Y%       | Z,ZZZ    |
| Group 3 | \$X.X    | Y%       | Z,ZZZ    |
| Group 4 | \$X.X    | Y%       | Z,ZZZ    |
| Group 5 | \$X.X    | Y%       | Z,ZZZ    |

# Choosing an effective visual

## Tables

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```
rows = [f'Category {i}' for i in range(1, 7)]  
cols = ['A', 'B', 'C']
```

```
fig, ax = plt.subplots(figsize=(5, 3.8))  
ax.axis('off')
```

```
table = ax.table(  
    cellText=data,  
    rowLabels=rows,  
    colLabels=cols,  
    cellLoc='center',  
    loc='center'  
)
```

```
plt.show()
```

# Choosing an effective visual Heatmaps

Table

|            | A   | B   | C   |
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Heatmap

LOW-HIGH

|            | A   | B   | C   |
|------------|-----|-----|-----|
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# Choosing an effective visual

## Heatmap Heatmaps

LOWHIGH

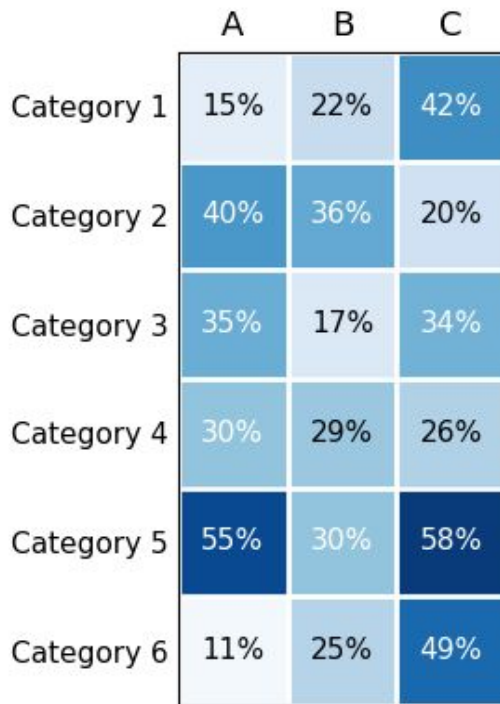
|            | A   | B   | C   |
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# Choosing an effective visual

## Heatmaps

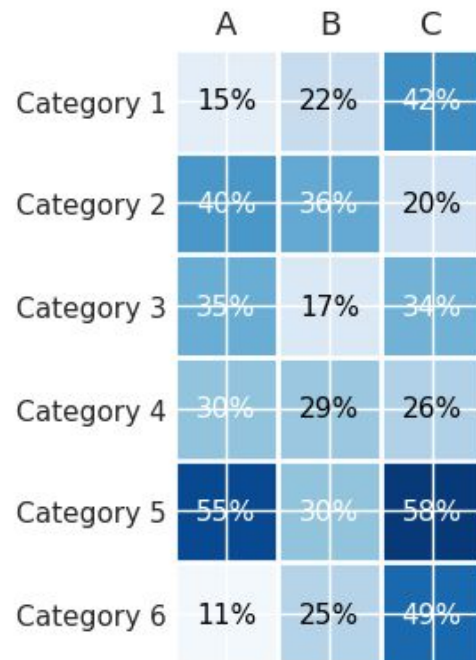
### Heatmap

LOWHIGH



### Heatmap

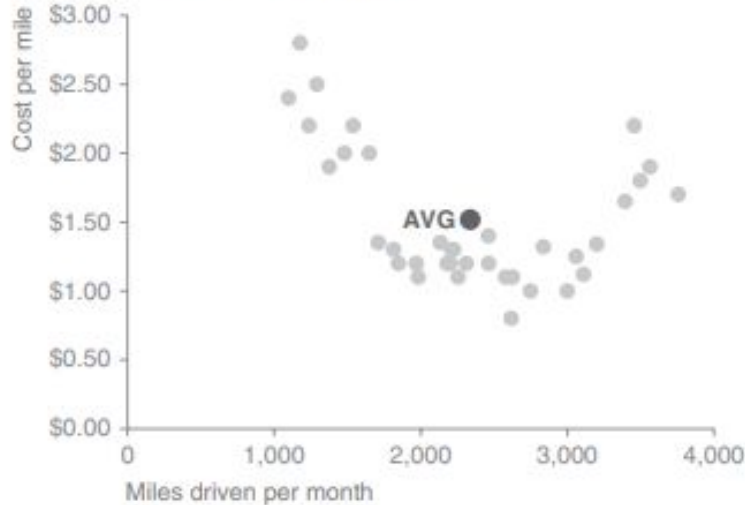
LOWHIGH



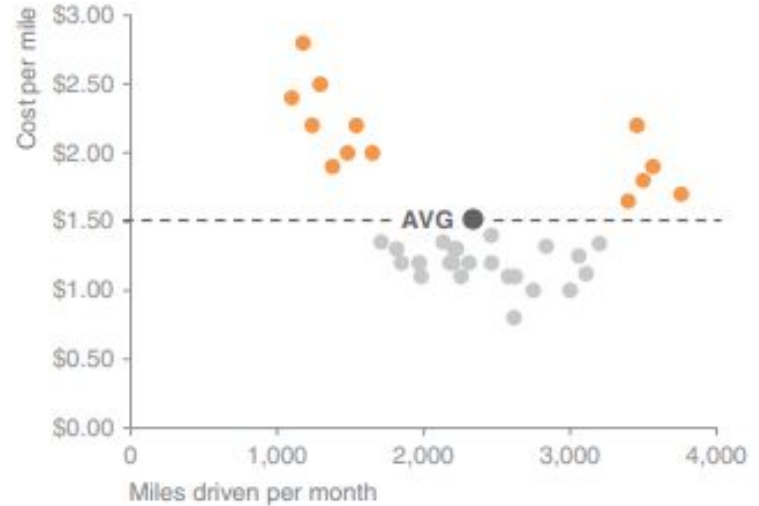
# Choosing an effective visual

## Graphs: Points (scatterplot)

Cost per mile by miles driven

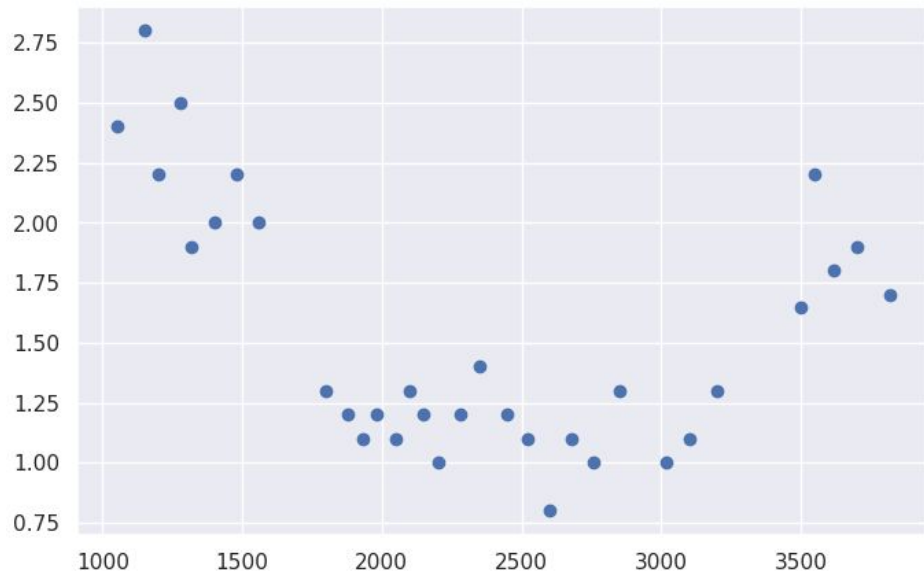


Cost per mile by miles driven





# Choosing an effective visual Graphs: Points (scatterplot)



```
x = np.array([ 1050,1150, ...  
])  
  
y = np.array([ 2.4,2.8, ...  
])  
  
fig, ax = plt.subplots(figsize=(8, 5))  
  
ax.scatter(  
    x,  
    y  
)  
  
plt.show()
```

# Choosing an effective visual

## Graphs: Lines

Single series



Two series



Multiple series



Passport control wait time

Past 13 months



# Choosing an effective visual

## Graphs: Slopegraph

Employee feedback over time



Employee feedback over time



# Choosing an effective visual

## Graphs: Bars

Too thin



Too thick



Just right

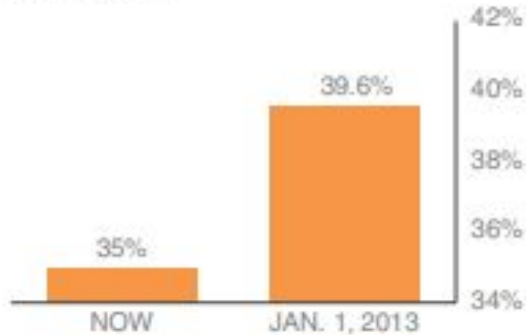


# Choosing an effective visual

## Graphs: Bars

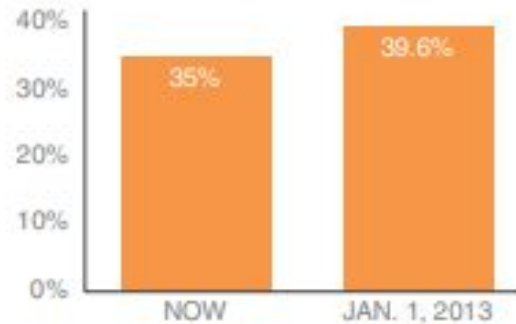
**Non-zero baseline:** as originally graphed

IF BUSH TAX CUTS EXPIRE  
TOP TAX RATE



**Zero baseline:** as it should be graphed

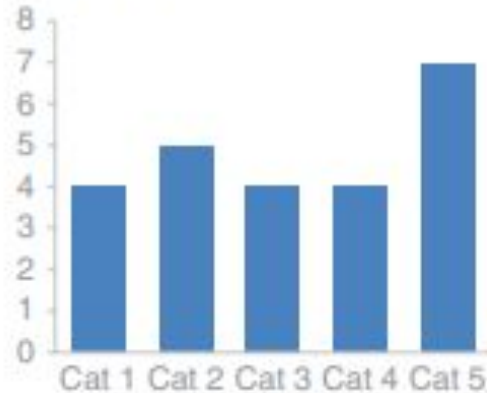
IF BUSH TAX CUTS EXPIRE  
TOP TAX RATE



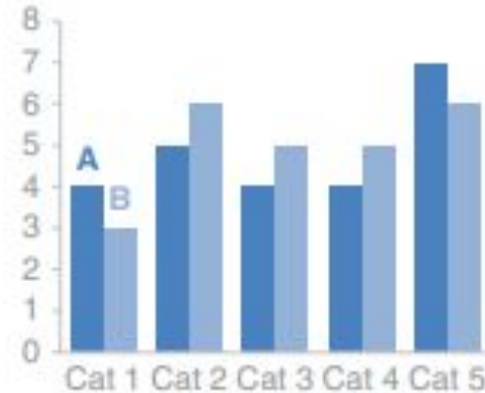
# Choosing an effective visual

## Graphs: Vertical Bar chart

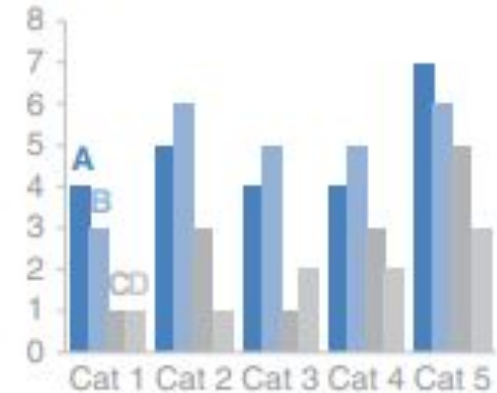
Single series



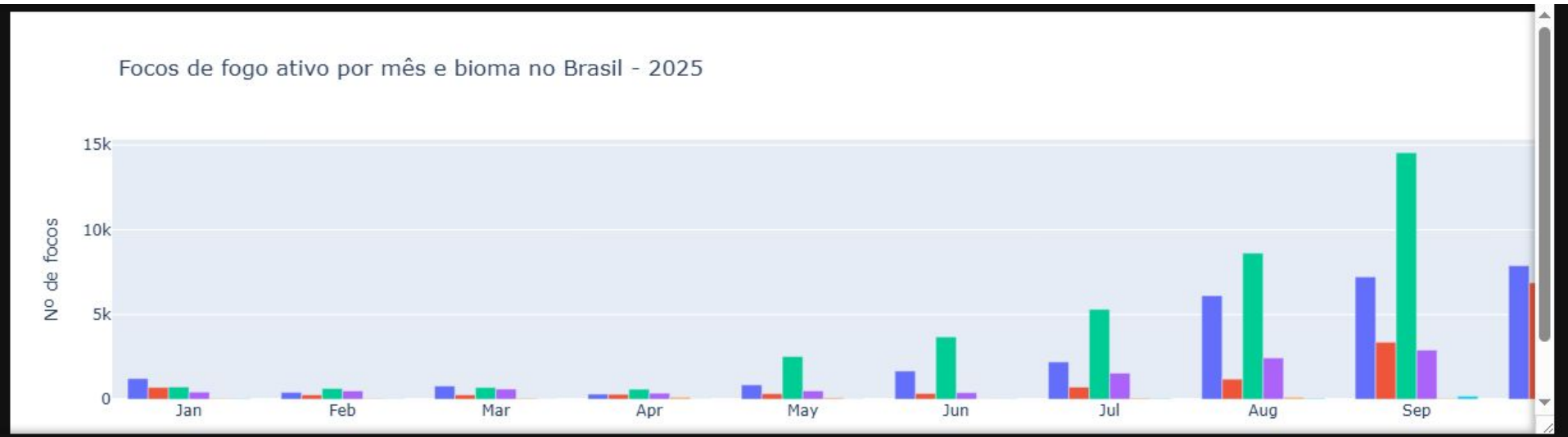
Two series



Multiple series



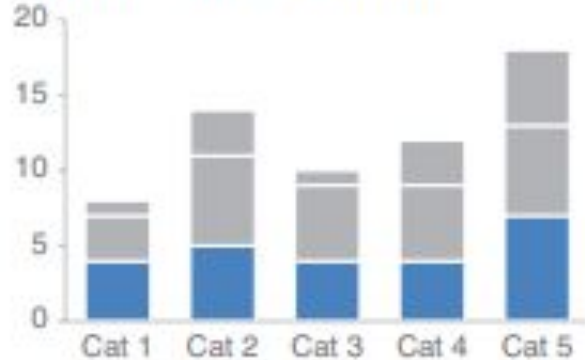
# Choosing an effective visual Graphs: Vertical Bar chart



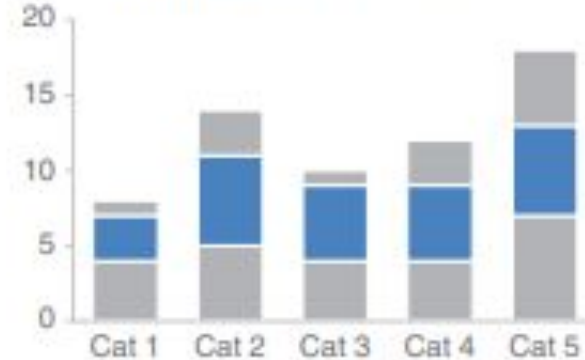
# Choosing an effective visual

## Graphs: Stacked Vertical Bar chart

Comparing **these** is easy



Comparing **these** is hard





# Choosing an effective visual Graphs: Waterfall chart

## 2014 Headcount math

Though more employees transferred out of the team than transferred in, aggressive hiring means overall headcount (HC) increased 16% over the course of the year.



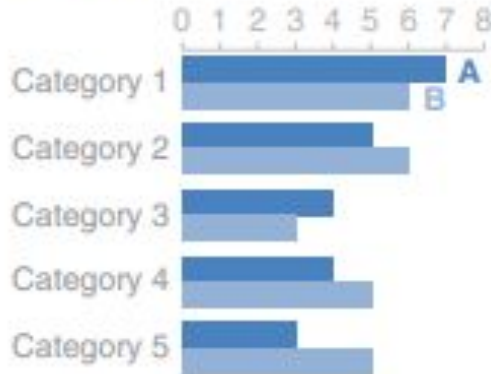
# Choosing an effective visual

## Graphs: Horizontal bar chart

Single series



Two series



Multiple series



# Choosing an effective visual

## Graphs: Stacked Horizontal bar chart

Survey results

Strongly Disagree | Disagree | Neutral | Agree | Strongly Agree

Percent of total

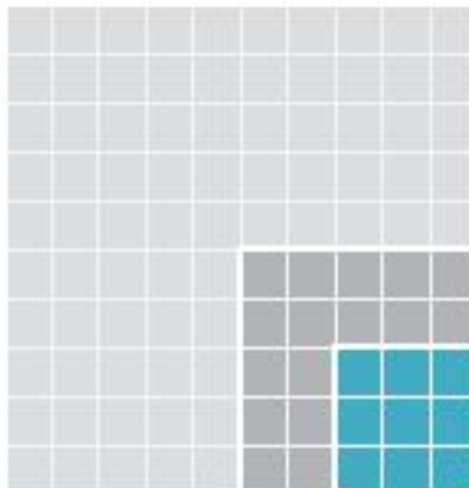
0% 20% 40% 60% 80% 100%



# Choosing an effective visual

## Graphs: Area

Interview breakdown



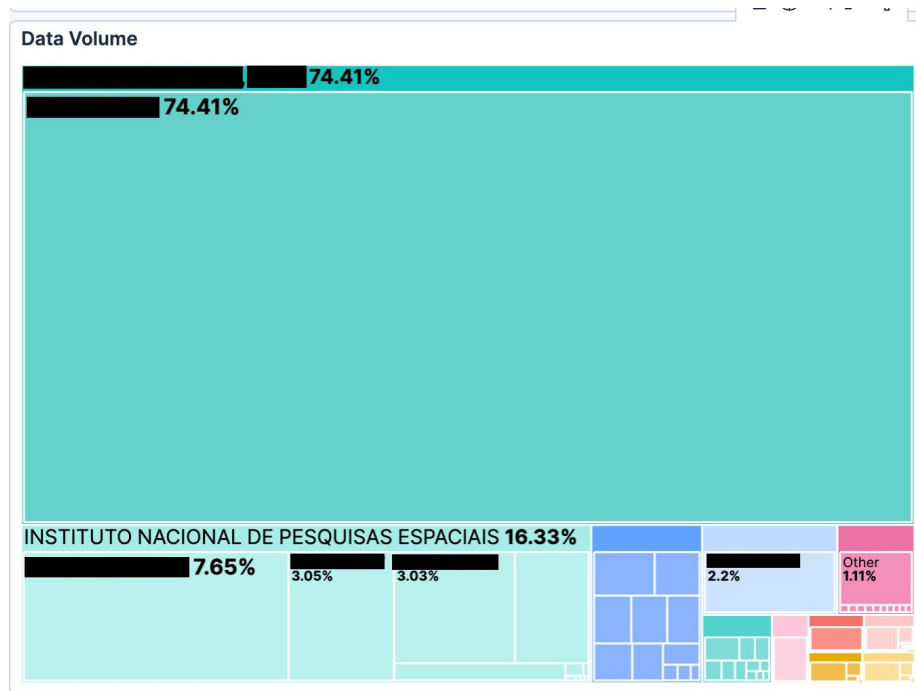
Out of every 100  
phone screens...

we bring 25  
**candidates onsite**  
for interviews...

and  
**extend 9 offers.**

# Choosing an effective visual

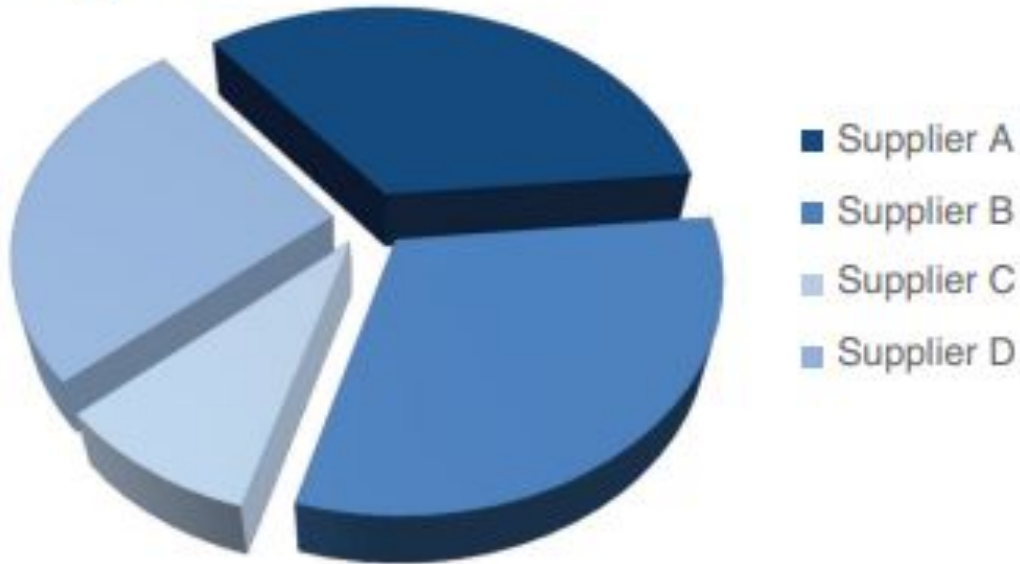
## Graphs: Area



# Choosing an effective visual

## Avoid

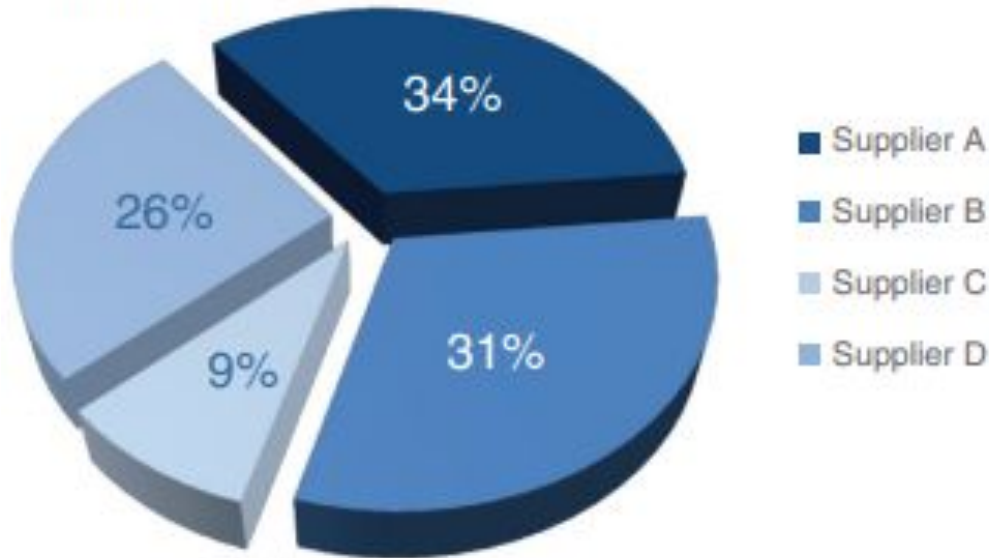
Supplier Market Share



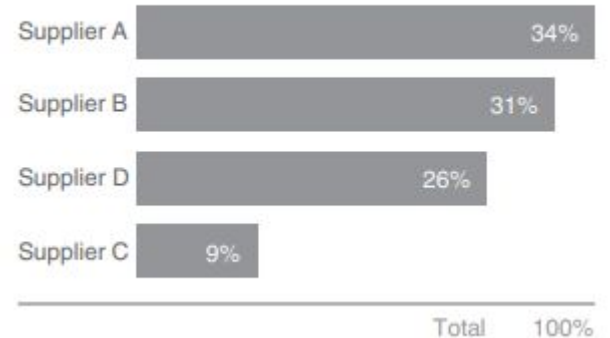
# Choosing an effective visual

## Avoid

Supplier Market Share



Supplier Market Share



# Choosing an effective visual

## Avoid

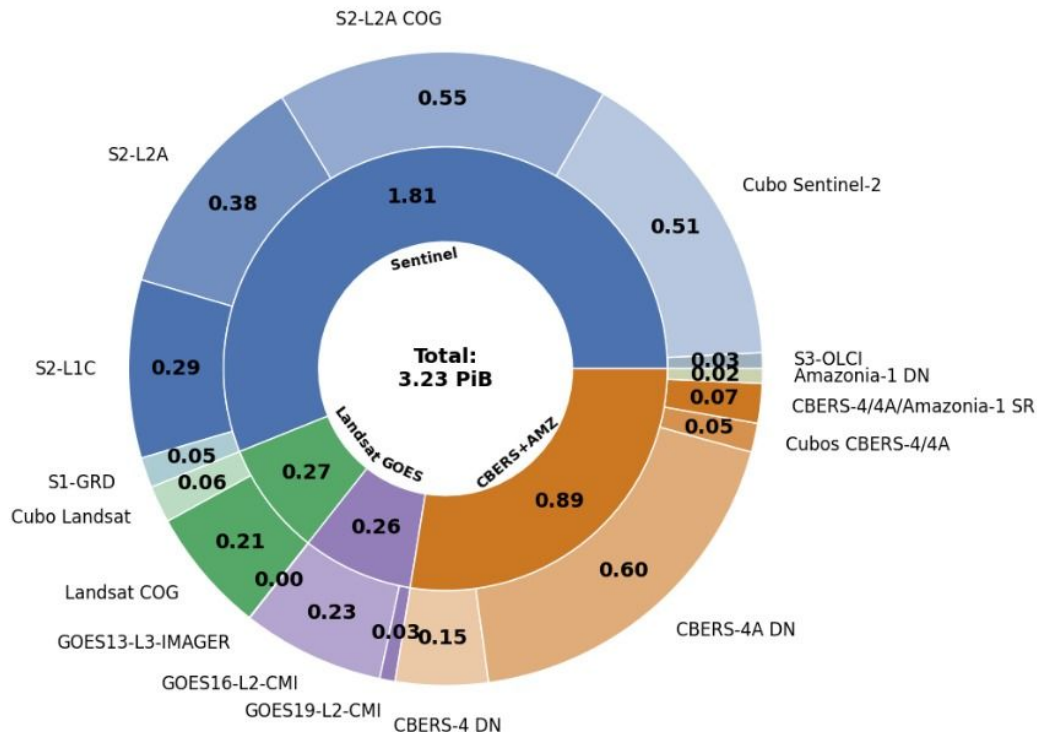
The donut chart





# Choosing an effective visual

**Volume de Dados Integrados e Disponibilizados na BIG  
(Consolidação até Fevereiro de 2026)**



# Choosing an effective visual

## Avoid

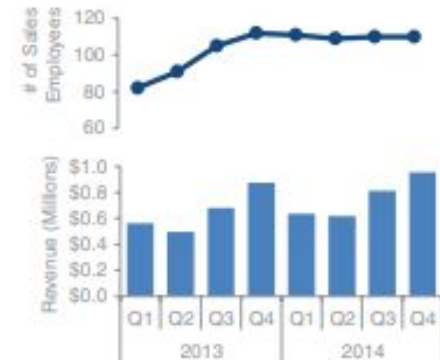
Secondary y-axis



Alternative 1: label directly



Alternative 2: pull apart vertically



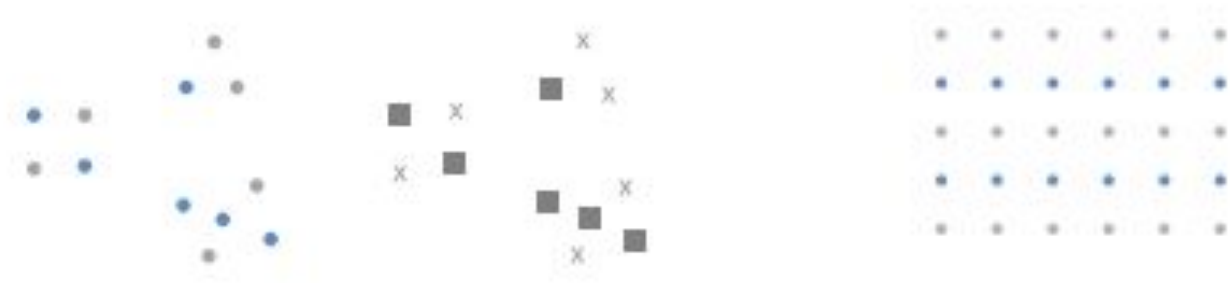


— **Clutter is  
your enemy**

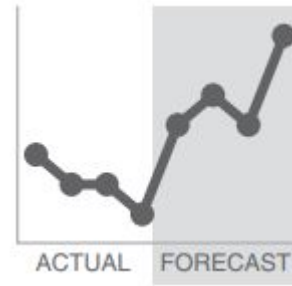
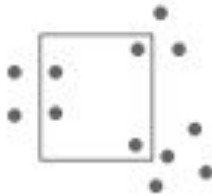
# Proximity



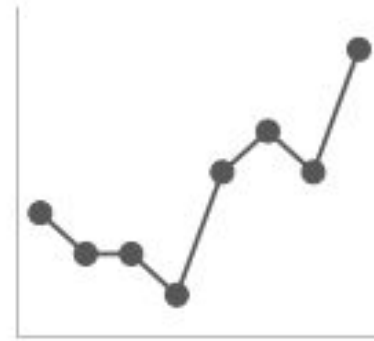
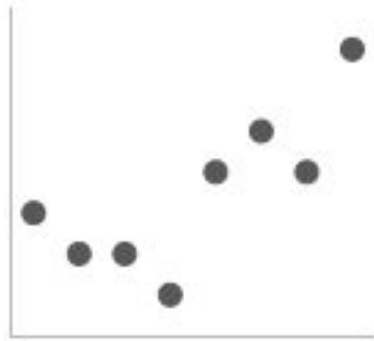
# Similarity



# Enclosure

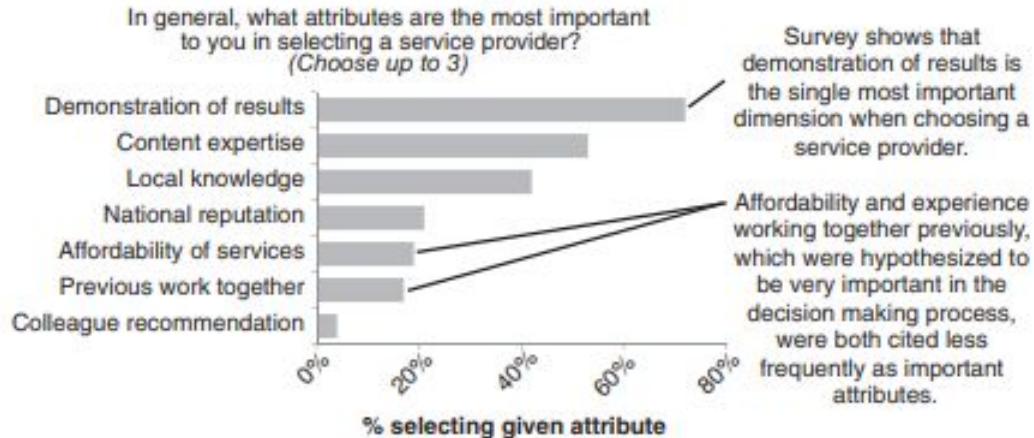


# Connection



# Lack of visual order

## Demonstrating effectiveness is most important consideration when selecting a provider



Data source: xyz; includes N number of survey respondents. Note that respondents were able to choose up to 3 options.

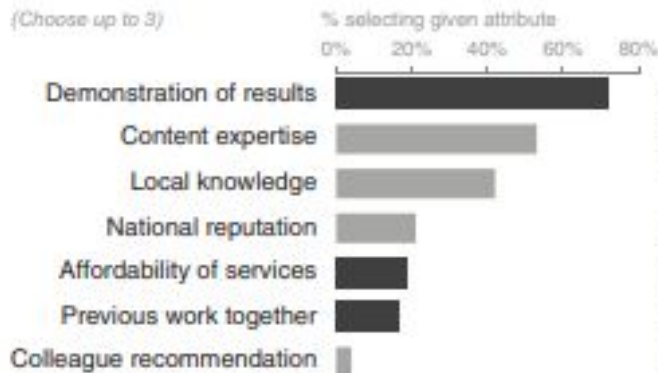


# Similarity

**Demonstrating effectiveness** is most important consideration when selecting a provider

In general, **what attributes** are the most important to you in selecting a service provider?

(Choose up to 3)



Survey shows that **demonstration of results** is the single most important dimension when choosing a service provider.

**Affordability** and **experience working together previously**, which were hypothesized to be very important in the decision making process, were both cited less frequently as important attributes.

Data source: xyz; includes N number of survey respondents.  
Note that respondents were able to choose up to 3 options.

# Contrast



# Contrast

## Performance overview

### ■ Our business

■ Competitor A

■ Competitor B

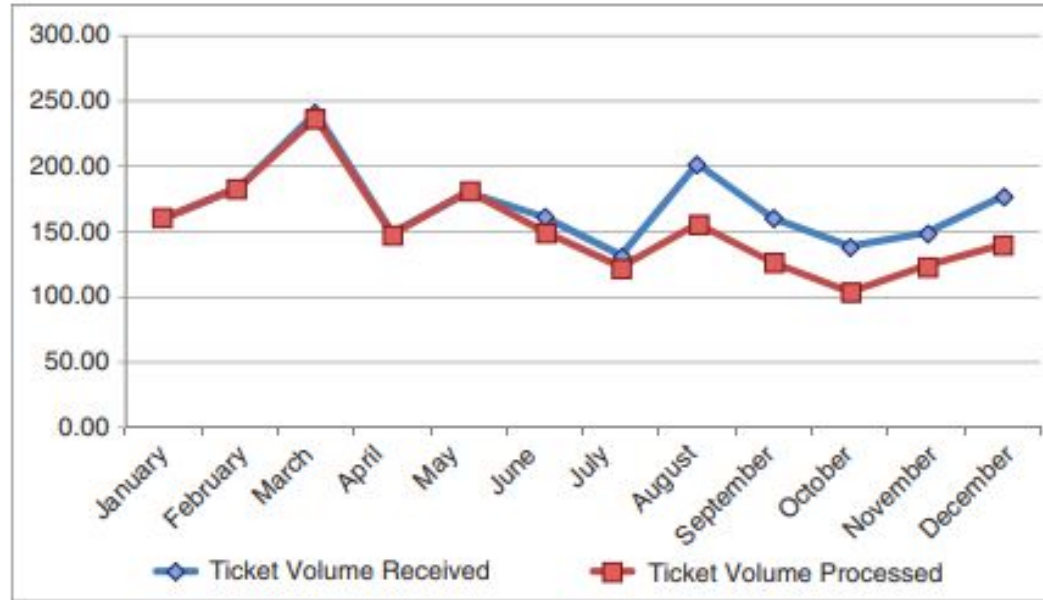
■ Competitor C

■ Competitor D

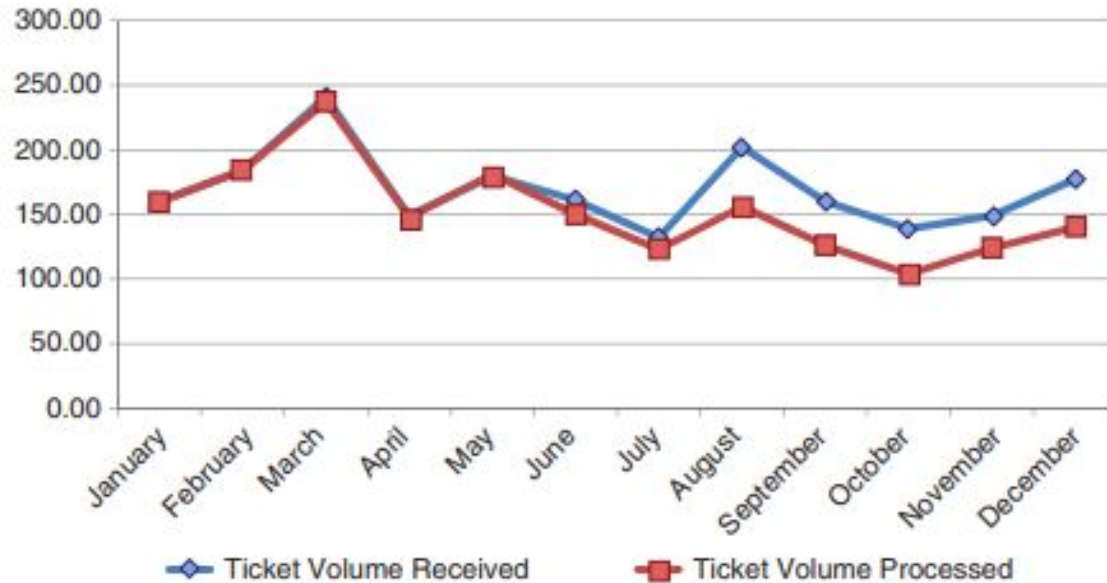
■ Competitor E



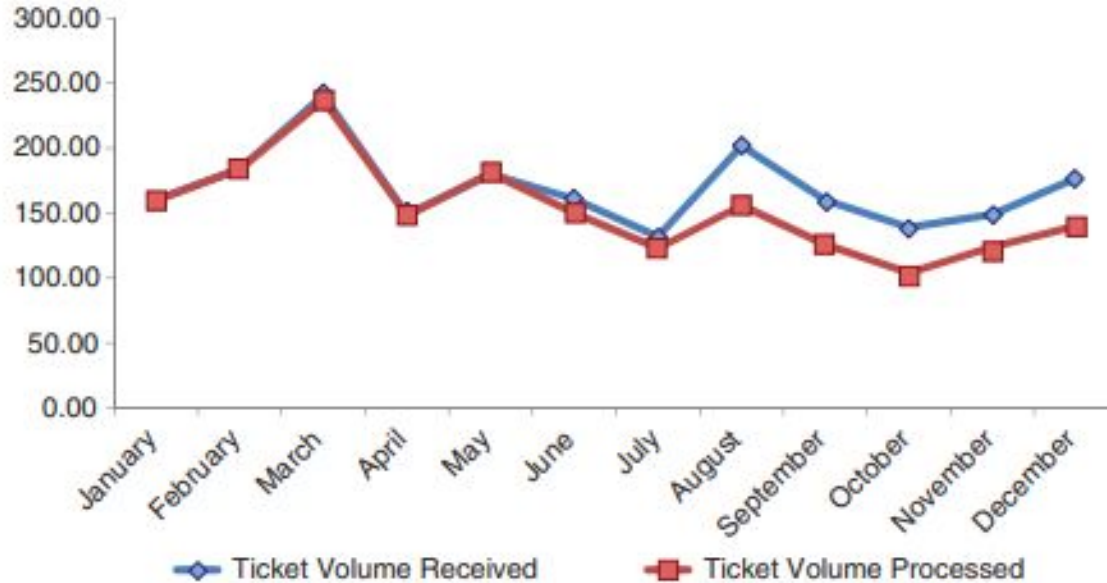
# Decluttering: step-by-step



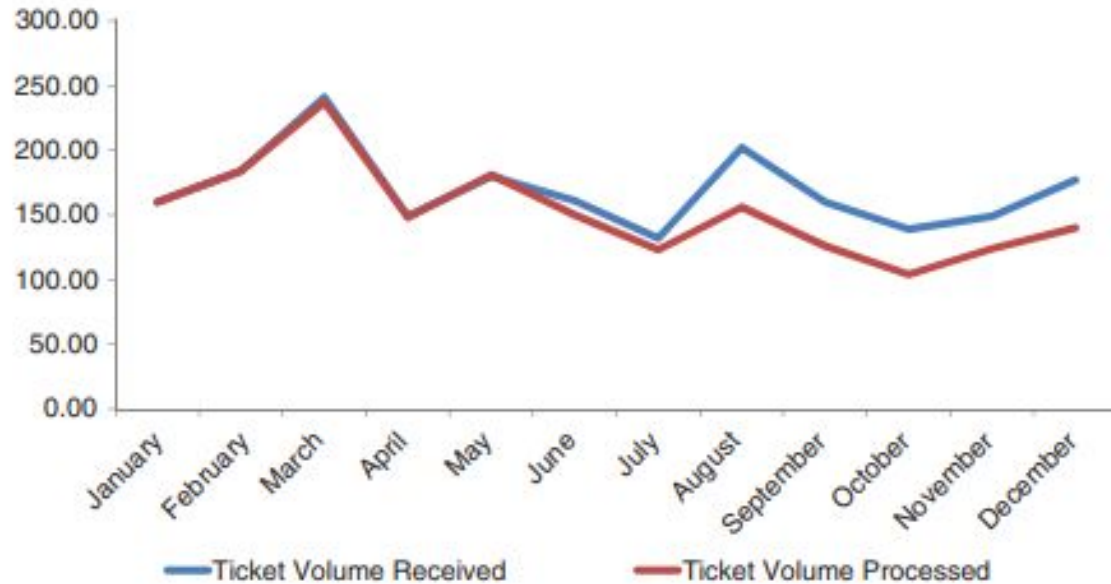
# Decluttering: step-by-step



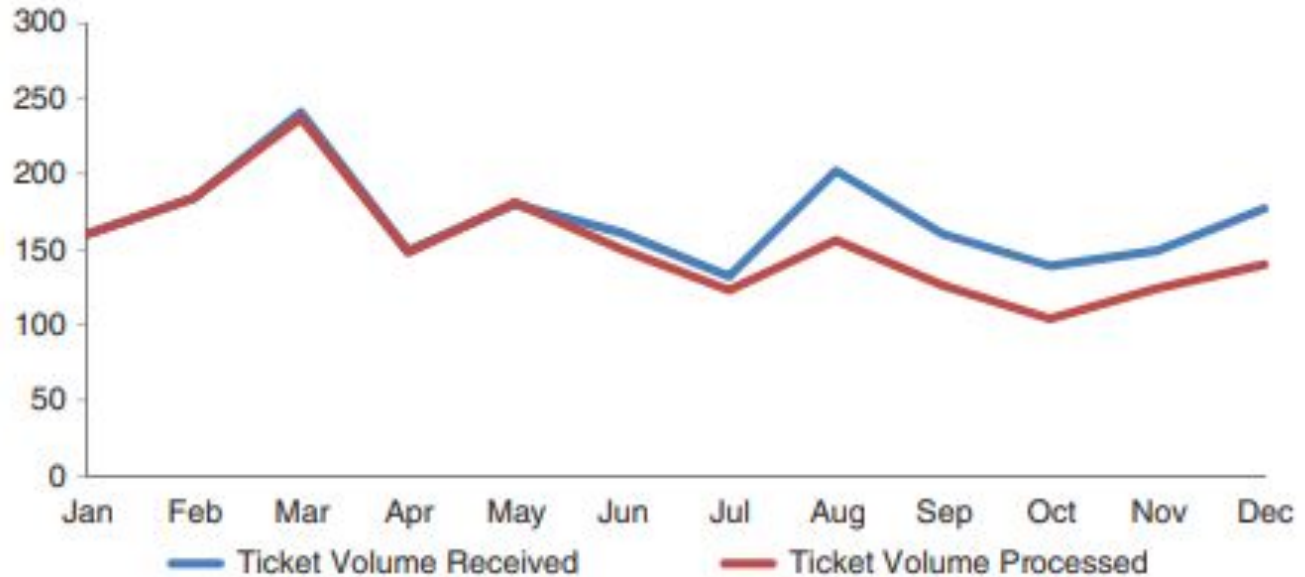
# Decluttering: step-by-step



# Decluttering: step-by-step

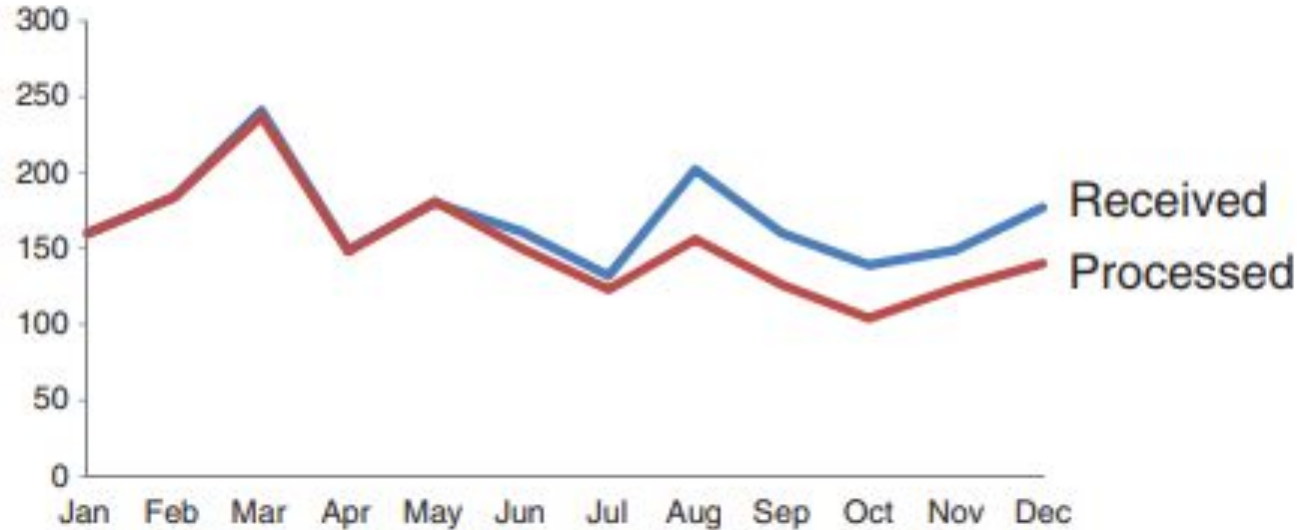


# Decluttering: step-by-step

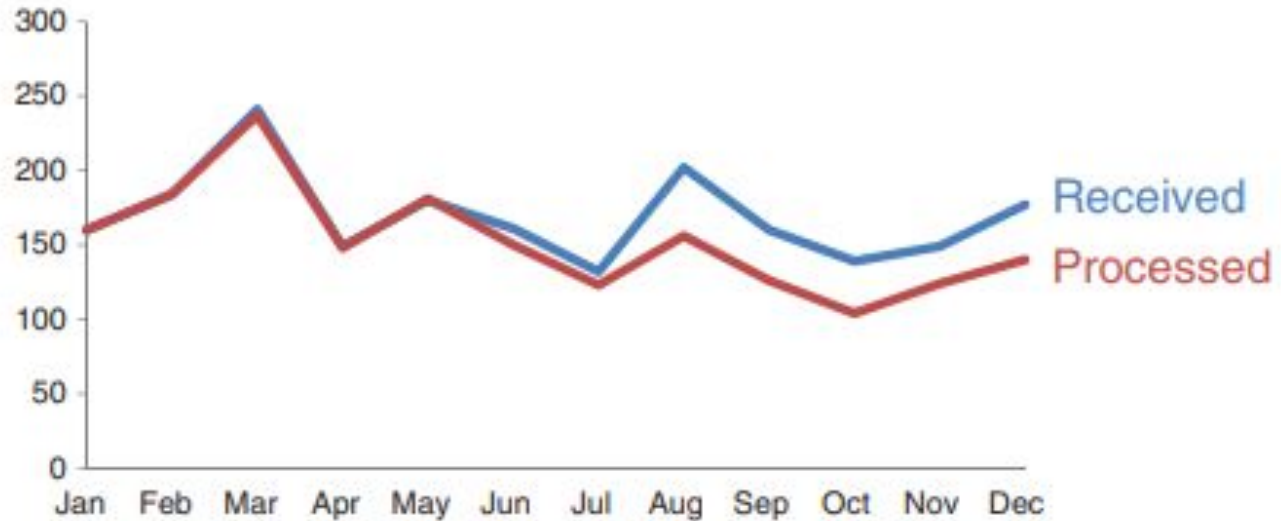




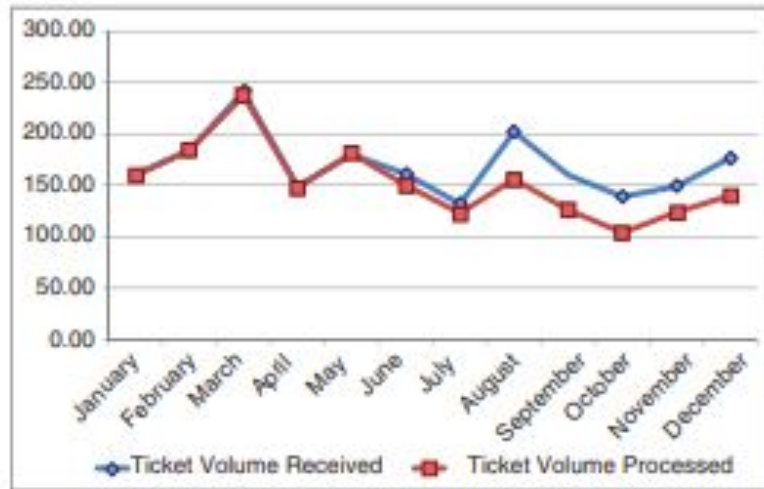
# Decluttering: step-by-step



# Decluttering: step-by-step

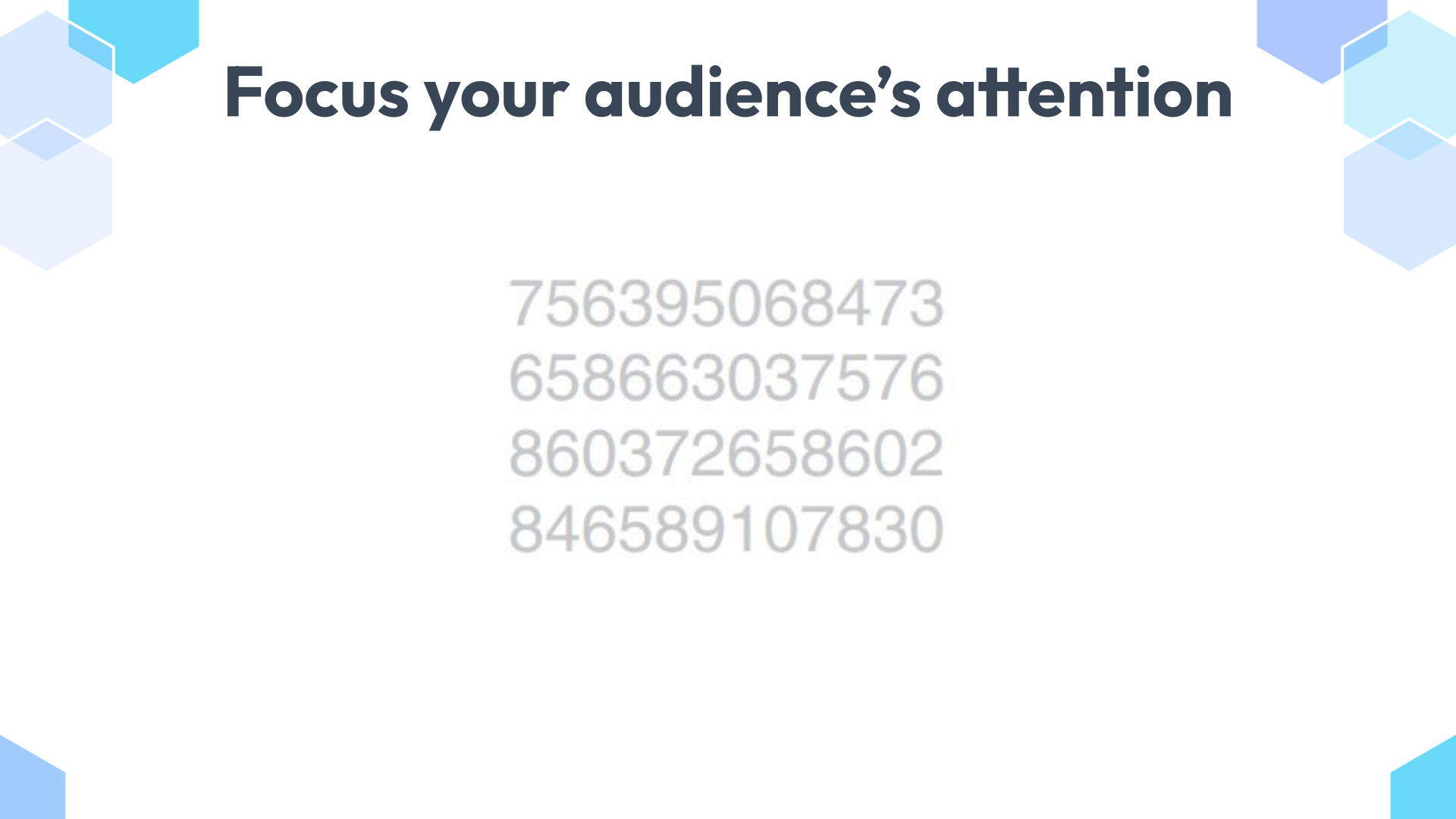


# Decluttering: step-by-step





—  
**Focus your  
audience's  
attention**



# Focus your audience's attention

756395068473

658663037576

860372658602

846589107830

# Focus your audience's attention

756**3**95068473

65866**303**7576

860**3**72658602

8465891078**30**

# Preattentive attributes



Orientation



Shape



Line length



Line width



Size



Curvature



Added marks



Enclosure



Hue



Intensity



Spatial position



Motion

# Preattentive attributes in text

## No preattentive attributes

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Color

What are we doing well? Great Products. **These products are clearly the best in their class.** Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Bold

**What are we doing well?** Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Italics

What are we doing well? Great Products. These products are clearly the best in their class. *Replacement parts are shipped when needed.* You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Size

What are we doing well? Great Products. These products are the best in their class. Replacement parts are shipped when needed. You sent gaskets **without me having to ask.** Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Outline (enclosure)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

## Separate spatially

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask.

Problems are resolved promptly.

Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!

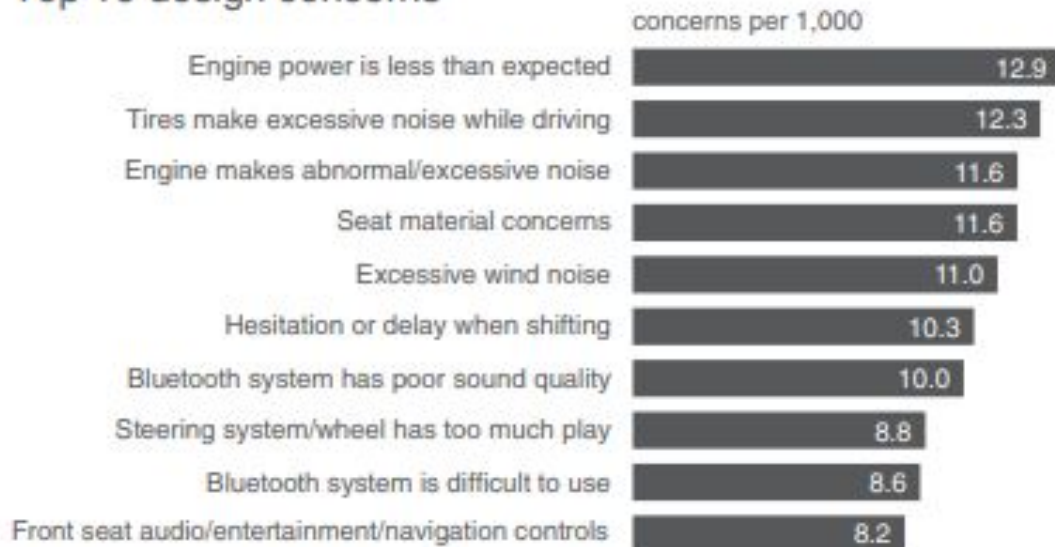
## Underline (added marks)

What are we doing well? Great Products. These products are clearly the best in their class. Replacement parts are shipped when needed. You sent me gaskets without me having to ask. Problems are resolved promptly. Bev in the billing office was quick to resolve a billing issue I had. General customer service exceeds expectations. The account manager even called to check in after normal business hours. You have a great company – keep up the good work!



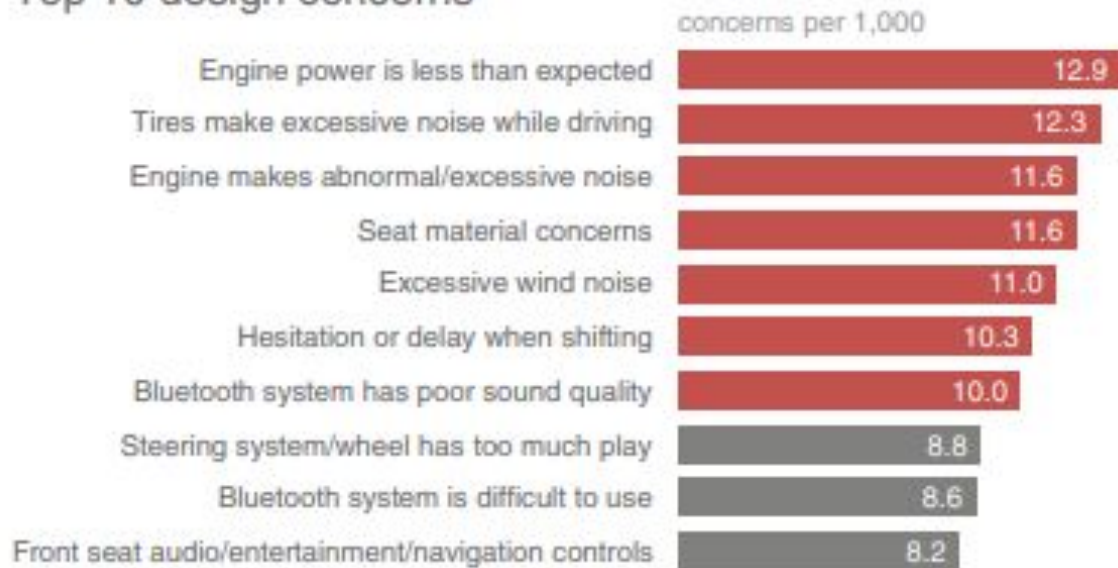
# Preattentive attributes in graphs

## Top 10 design concerns



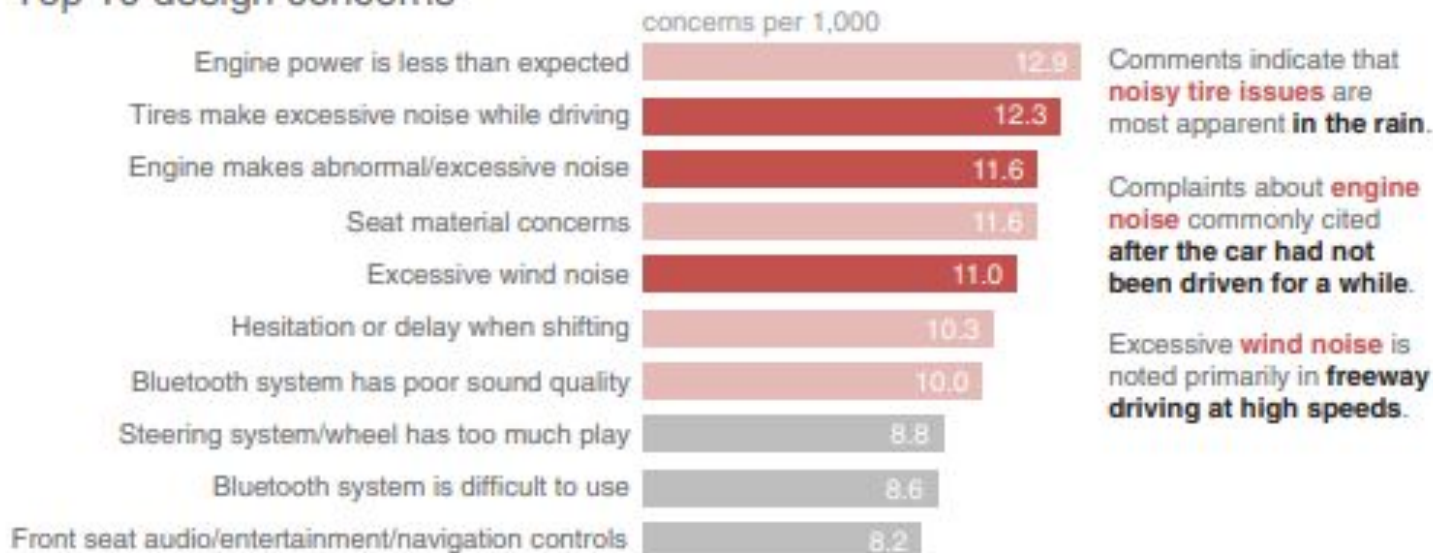
# Preattentive attributes in graphs

## Top 10 design concerns

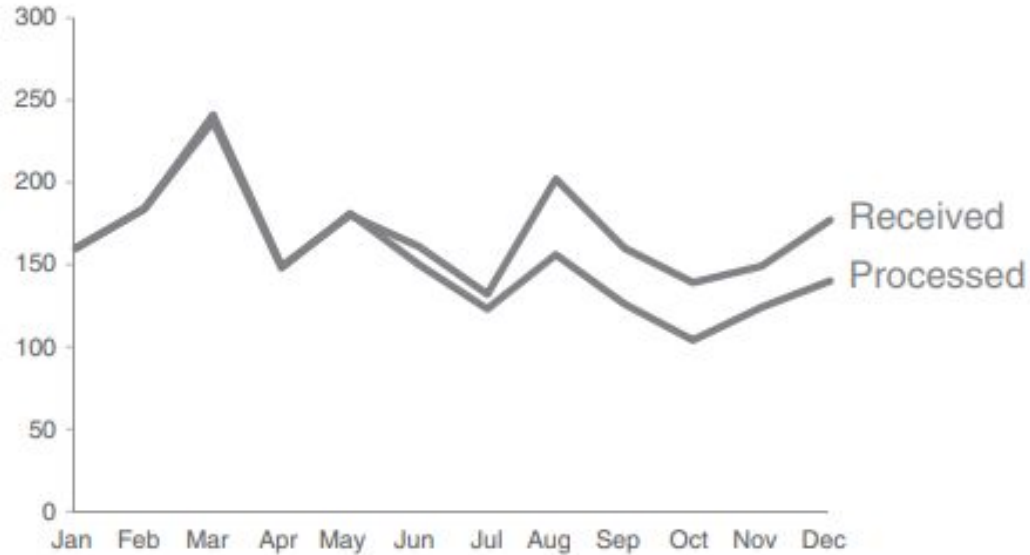


# Preattentive attributes in graphs

## Top 10 design concerns



# Preattentive attributes in graphs



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# Preattentive attributes in graphs

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

| Country | A | B | C  | D  | E  |
|---------|---|---|----|----|----|
| AUS     | 1 | 2 | 3  | 6  | 7  |
| BRA     | 1 | 3 | 4  | 5  | 6  |
| CAN     | 2 | 3 | 6  | 12 | 8  |
| CHI     | 1 | 2 | 8  | 4  | 7  |
| FRA     | 3 | 2 | 4  | 8  | 10 |
| GER     | 3 | 1 | 6  | 5  | 4  |
| IND     | 4 | 1 | 8  | 10 | 5  |
| ITA     | 2 | 4 | 10 | 9  | 8  |
| MEX     | 1 | 5 | 4  | 6  | 3  |
| RUS     | 4 | 3 | 7  | 9  | 12 |
| SPA     | 2 | 3 | 4  | 5  | 11 |
| TUR     | 7 | 2 | 3  | 4  | 8  |
| UK      | 1 | 2 | 3  | 6  | 7  |
| US      | 1 | 2 | 4  | 3  | 5  |

Top 5 drugs: country-level sales rank

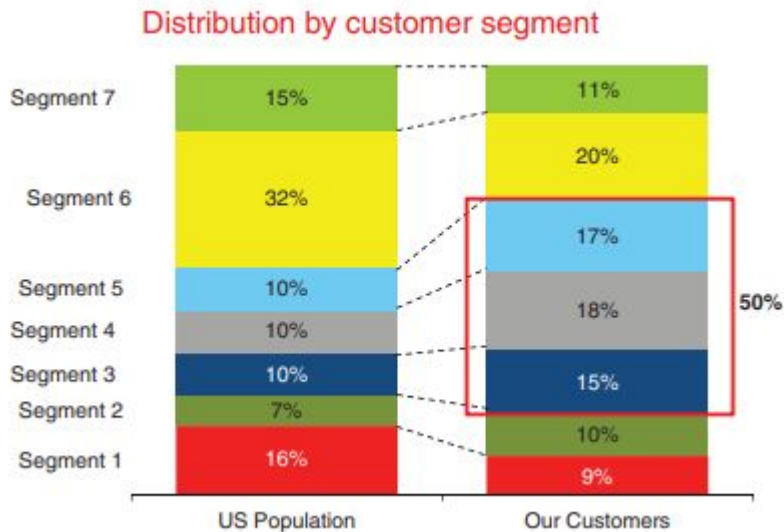
RANK 1 2 3 4 5+

| COUNTRY   DRUG |  | A | B | C  | D  | E  |
|----------------|--|---|---|----|----|----|
| Australia      |  | 1 | 2 | 3  | 6  | 7  |
| Brazil         |  | 1 | 3 | 4  | 5  | 6  |
| Canada         |  | 2 | 3 | 6  | 12 | 8  |
| China          |  | 1 | 2 | 8  | 4  | 7  |
| France         |  | 3 | 2 | 4  | 8  | 10 |
| Germany        |  | 3 | 1 | 6  | 5  | 4  |
| India          |  | 4 | 1 | 8  | 10 | 5  |
| Italy          |  | 2 | 4 | 10 | 9  | 8  |
| Mexico         |  | 1 | 5 | 4  | 6  | 3  |
| Russia         |  | 4 | 3 | 7  | 9  | 12 |
| Spain          |  | 2 | 3 | 4  | 5  | 11 |
| Turkey         |  | 7 | 2 | 3  | 4  | 8  |
| United Kingdom |  | 1 | 2 | 3  | 6  | 7  |
| United States  |  | 1 | 2 | 4  | 3  | 5  |

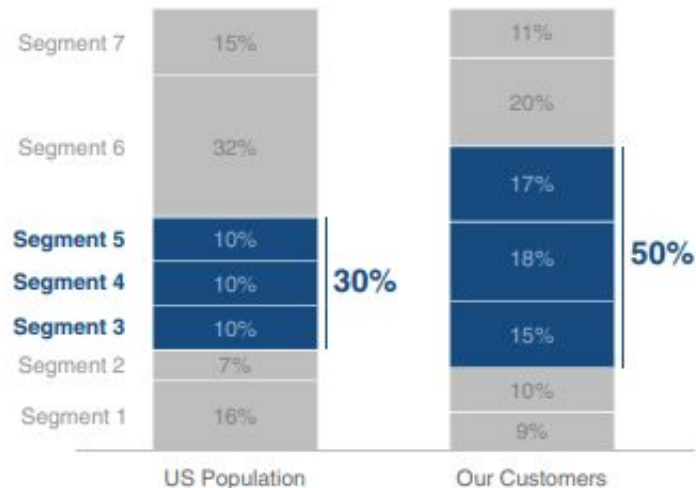
colorblind



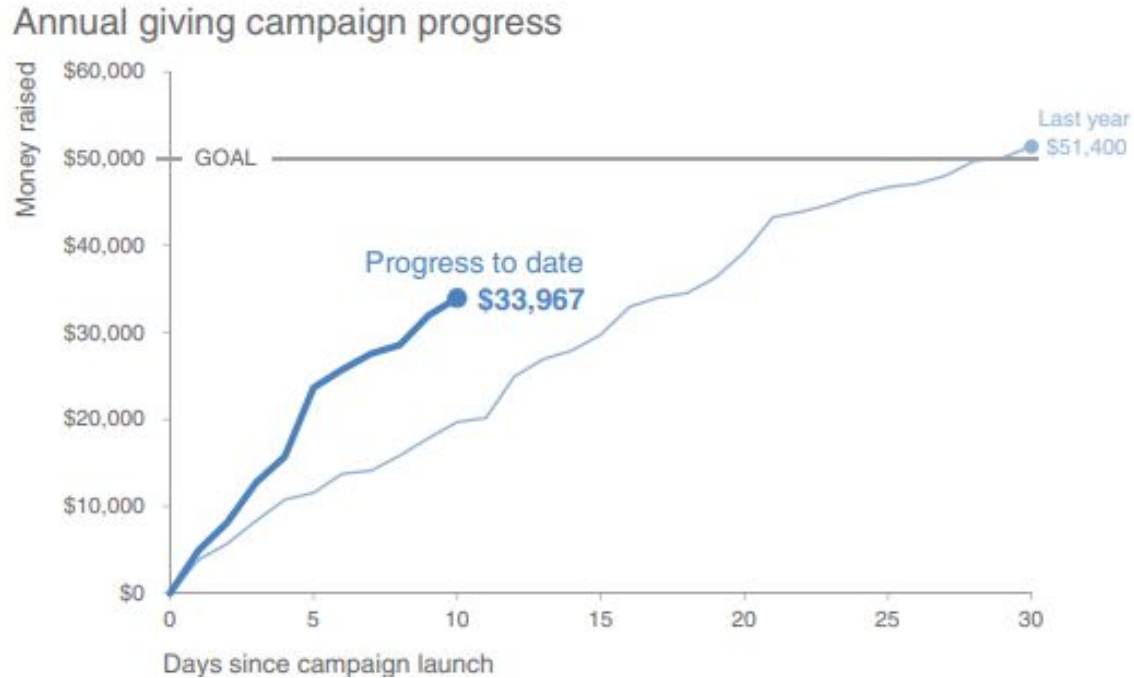
# Preattentive attributes in graphs



Distribution by customer segment



# What is this graph informing?



# What is this graph informing?

Sales over time

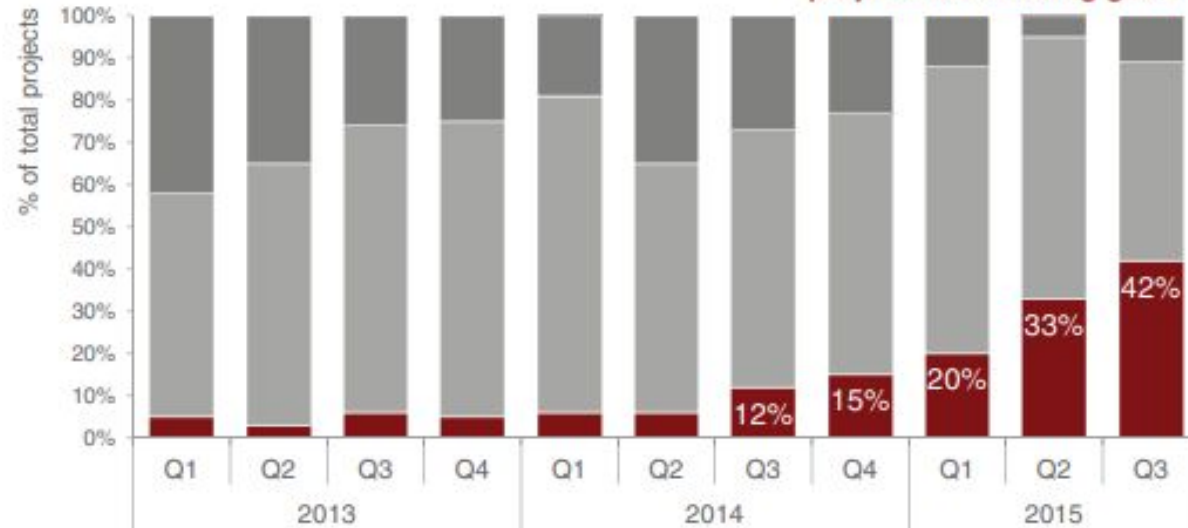


# What is this graph informing?

Goal attainment over time

■ Miss ■ Meet ■ Exceed

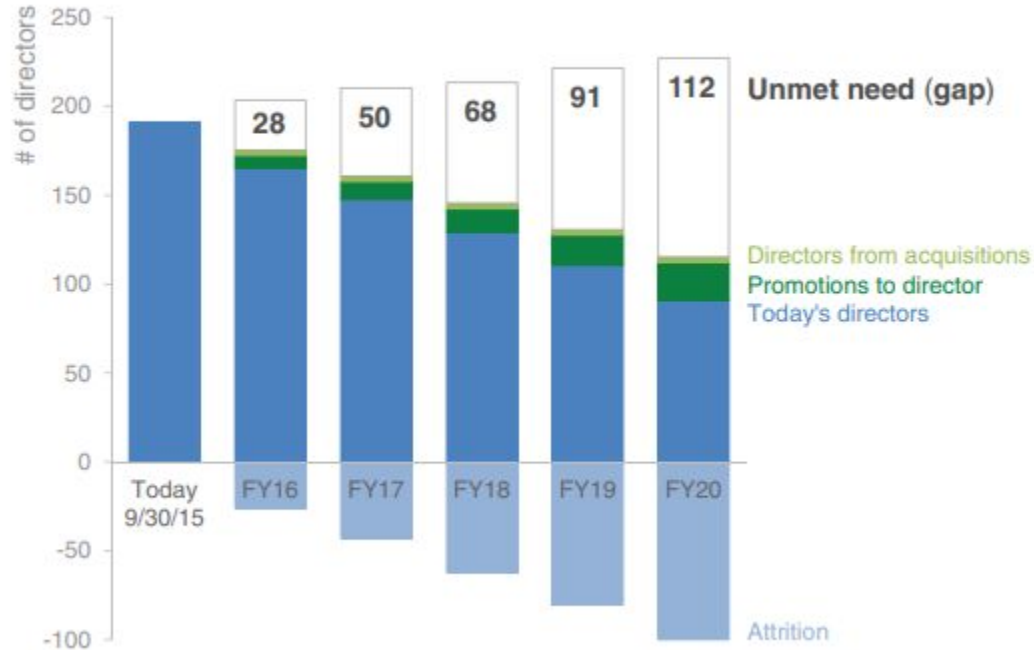
As of Q3 2015, more than 1/3 of projects are missing goals



Data source: XYZ Dashboard; the total number of projects has increased over time from 230 in early 2013 to nearly 270 in Q3 2015.

# What is this graph informing?

Expected director population over time



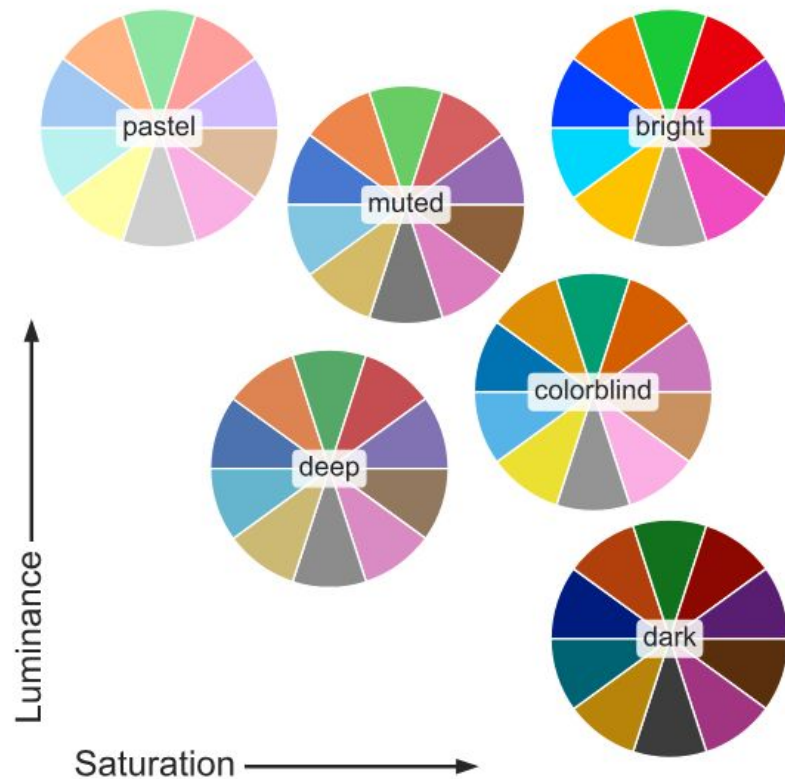
A footnote explaining relevant forecast assumptions and methodology would go here.



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# Colors

# Colors



```
sns.color_palette("flare", as_cmap=True)
```



```
sns.color_palette("crest", as_cmap=True)
```





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# Storytelling



# Storytelling

- The beginning
- The middle
- The end

# Storytelling

- The beginning
- The middle
- The end

repetition

# Review

- Context
- Choose appropriate display
- Eliminate clutter
- Draw attention where you want your audience to focus



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*matplotlib*





seaborn



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# Data Visualization

Rennan Marujo

